9 **(a)** Fig. 9.1 represents part of the emission spectrum of atomic hydrogen. It contains a series of lines and the wavelength of some of which are indicated. There are no lines in the series with wavelengths less than 91.2 nm.

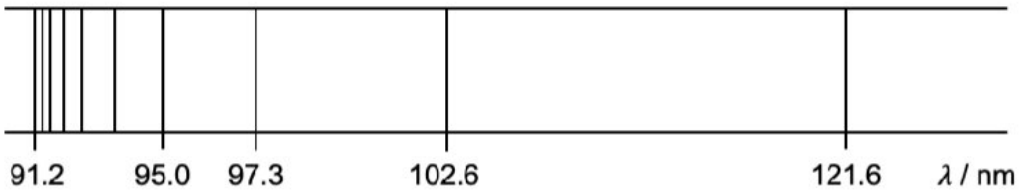


Fig. 9.1

(i) Show that the energy E of a photon, in eV, and its wavelength λ, in nm, are related by the equation:

$$E\lambda = 1243 \text{ eV nm}$$

[2]

(ii) The photon energies of the 91.2 nm, 102.6 nm and 121.6 nm lines are 13.6 eV, 12.1 eV and 10.2 eV respectively.

Determine the photon energies of the remaining two wavelengths indicated in Fig. 9.1.

photon energy of 95.0 nm = ... eV

photon energy of 97.3 nm = ... eV

[2]

(iii) The highest energy level for hydrogen is 0 eV.

The wavelengths in Fig. 9.1 are produced when the atom de-excite from higher energy to the ground state.

Use the values in **(a)(ii)** to construct the partial energy level diagram of the atom in Fig. 9.2.

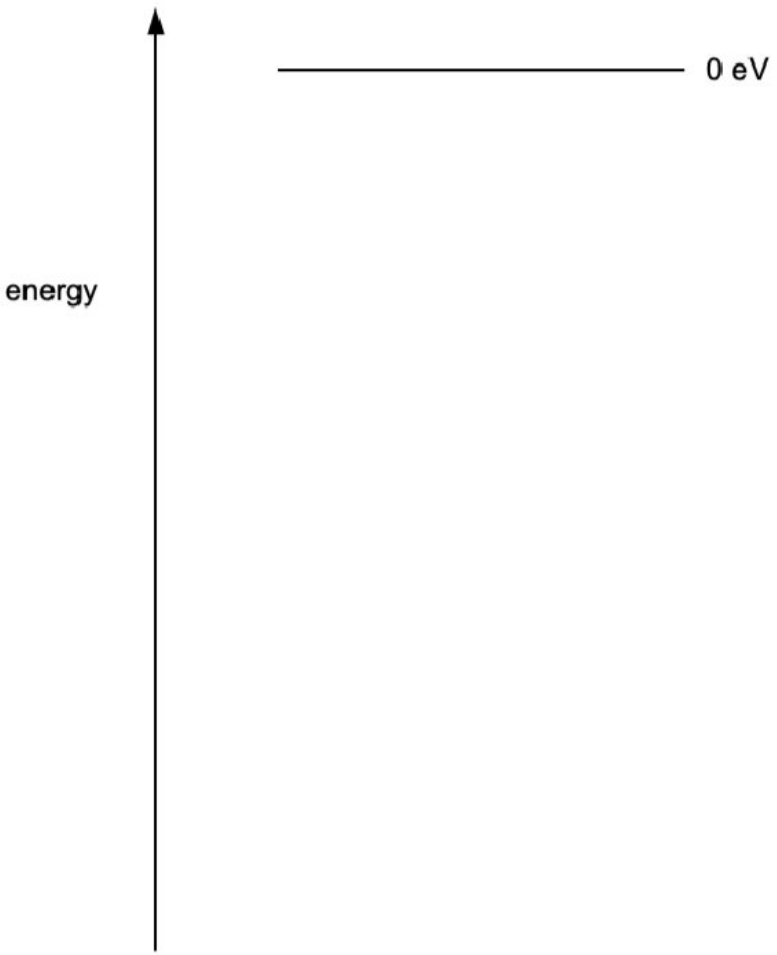


Fig. 9.2

[5]

(iv) Another line in the hydrogen occurs at a wavelength of 430 nm. Label on Fig. 9.2 the likely transition responsible for this line. [1]

(b) Fig. 9.3 shows two allowed wavefunctions ψ_A and ψ_B for an electron confined in a one-dimensional infinite square potential well of length L.

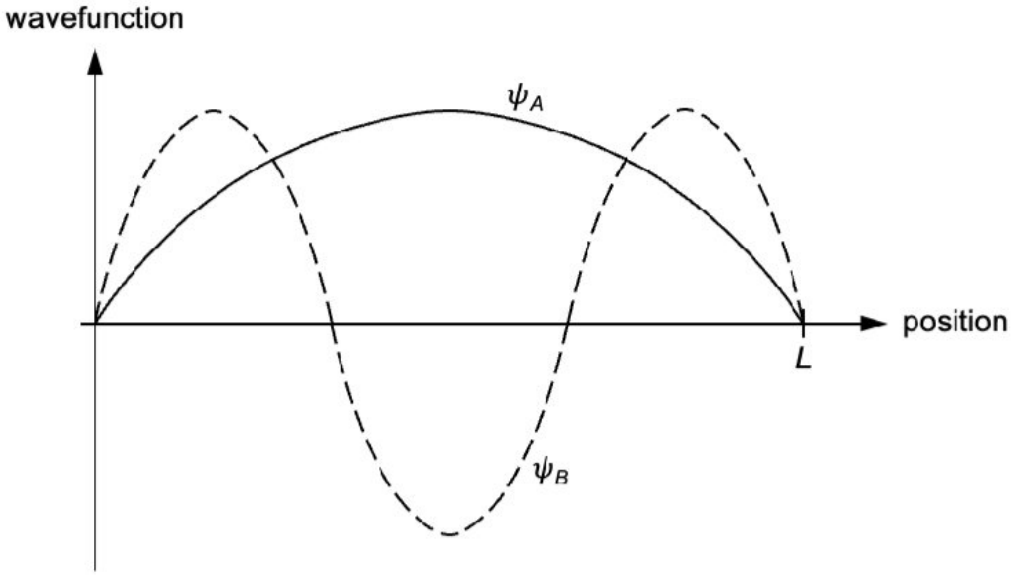


Fig. 9.3

(i) Explain which wavefunction corresponds to a higher electron energy level.

.....

.....

.....

.....

..... [3]

(ii) L is 1.0×10^{-10} m.

Determine the energy of the photon emitted when the electron transits from the higher energy level to the lower energy level.

energy of photon = ... J [3]

(c) A beam of electrons is directed along a normal towards a barrier, as shown in Fig. 9.4

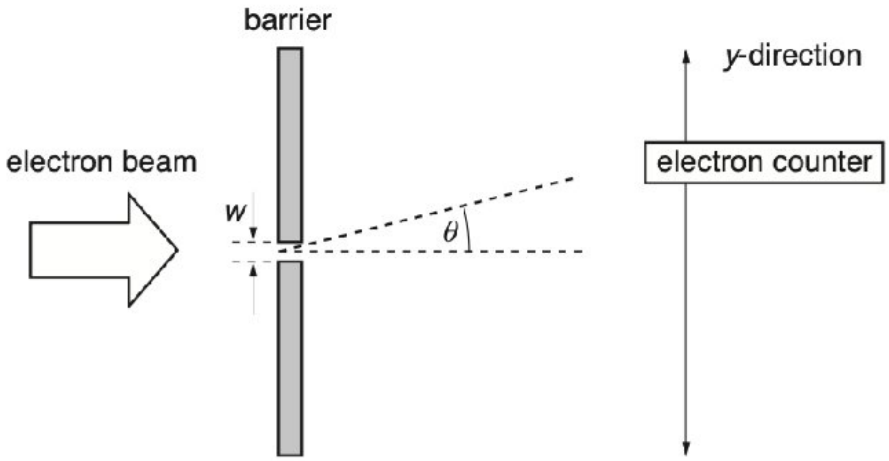


Fig. 9.4

The barrier contains a single slit of width *w*. Beyond the slit there is a detector that counts electrons. This can be moved in the *y*-direction to compare the rate of arrival of electrons at different values of the angle θ from the original direction of the beam.

Louis de Broglie suggested that some aspects of the behaviour of electrons can be explained using a wave model. The diffraction of the matter wave explains why the electron beam spreads out beyond the slit.

Werner Heisenberg used a different approach involving what is now known as the uncertainty principle. This can also be used to explain why the electron beam spreads out after passing through the slit.

(i) Calculate the uncertainty in momentum when an electron travelling at 2.8×10^{7} m s^{-1} towards the barrier passes through a narrow slit of width 1.5×10^{-10} m

uncertainty in momentum = ... N s [2]

(ii) Use the uncertainty principle to explain why the electron beam spreads out beyond the slit.

.....

.....

.....

..... [2]

[Total: 20]